Machine Learning Models for Sentiment Analysis: State of the Art and Applications

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2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT) | 979-8-3503-7024-9/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCCNT61001.2024.10726197

***Abstract*— Sentiment Analysis (SA) is one of the prominent and emerging research area in the domain of natural language processing. This paper explores the field of sentiment analysis using machine learning (ML) models. It aims to provide a comprehensive overview of the current state of the art, applications, limitations, and future directions of SA. It identifies various ML models like Random Forest, Logistic Regression, Decision Tree, Support Vector Machine, and Naïve Bayes used for SA. It also discusses various applications of SA across different domains like product reviews, social media monitoring, financial forecasting, recommender systems and crisis management. While acknowledging the limitations of SA like linguistic barriers, sarcasm detection, and noisy data, the paper proposes future research directions to address these challenges and improve the accuracy and efficiency of SA models.**

Keywords—Machine Learning, Sentiment Analysis, Natural Language Processing, Deep Learning.

I. INTRODUCTION

As the world undergoes modernization, new technological breakthroughs are a regular occurrence. The internet facilitates the rapid dissemination of news about these advancements globally within seconds. Consequently, information has become accessible to individuals worldwide. This unrestricted access to vast amounts of data has given rise to various perspectives and opinions, leading to the emergence of blogs, articles, online forums, and social networks as platforms for open dialogue and expression on diverse topics. Sentiment analysis (SA), also known as opinion mining or mood detection, is experiencing significant growth and relevance in the field of Natural Language Processing (NLP). It aims to learn and discern thoughts, emotions, and feelings from textual content[1]. With the abundance of user-generated content on the World Wide Web, including articles and review sites, coupled with the popularity of social media platforms like Twitter and YouTube, SA has become a sought-after tool for analyzing community feedback, particularly for businesses and global research. This paper delves into the intriguing research area of sentiment analysis and seeks to find novel insights to the theory, methods, and applications within this field. The ability to interpret content and detect sentiment is crucial in various aspects. For businesses, understanding customer reviews is influential for growth, and sentiment analysis can gauge a product's success in the market. Public opinion holds significance in politics, finance, and even in education, where analyzing online discourse becomes a pressing issue to understand public preferences and demands. Ongoing research reveals that Automatic Sentiment Analysis has certain limitations, making it a growing research topic [2]. Despite numerous applications, sentiment analysis faces conceptual and applied constraints that impede comprehensive accuracy. For instance, a study by Thangaraju et al. [3] proposed an Ensemble learning-based sentiment analysis for diverse content, utilizing GAN and SAN techniques. However, limitations were observed in extracting meaningful information from short texts and analysing texts with mixed languages. Despite these limitations, sentiment analysis has benefited significantly from research endeavours. Therefore, to develop better models for SA, it is needed to investigate various machine learning models for sentiment analysis, understand the issues and challenges associated with these and eliminate these issues and challenges. Figure 1 shows the general framework of sentiment analysis using machine learning.

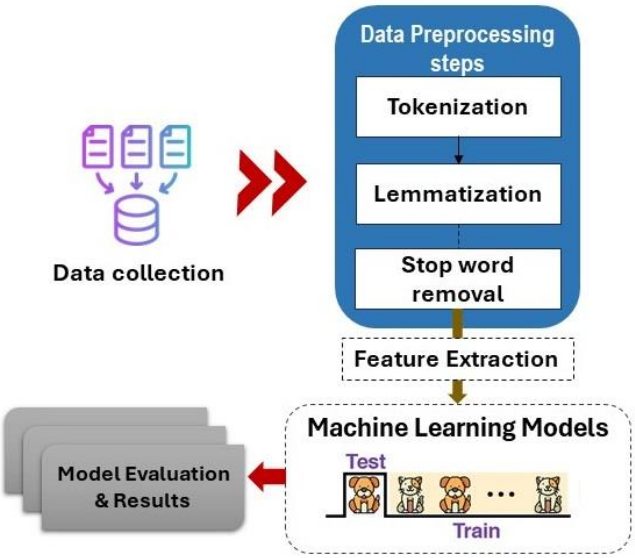


Fig 1. Framework of sentiment analysis using machine learning.

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Many researchers have made significant contributions by conducting thorough investigations, documenting their findings, and offering general comparisons of various SA approaches, their work is not without limitations, as outlined in Table 1. For example, the systematic literature review (SLR) conducted by Q. A. Xu et al. [4] covers aspects such as defined tools, machine learning models, evaluation metrics, and future research prospects but lacks a focus on SA applications and associated open issues. Similarly, paper by J. Abate et al. [5] and S. Alyamiet al. [6] also neglect these crucial areas. The study conducted by M. Subramanianet al. [7] lacks a detailed description of the tools used, as well as evaluation metrics and open issues. While the review by M. Obaidi et al. [1] is well-informed, but it does not discuss the evaluation metrics employed. Similarly, in papers [8] and [9], the authors omit addressing open issues in sentiment analysis, and [10] fails to provide explicit or detailed applications of the field. Furthermore, paper [11] overlooks significant areas, including tools used, applications, open issues, and future research directions. By comprehensively addressing these minor yet critical research aspects and integrating them into the broader academic discourse, this paper stands as a well-defined review on sentiment analysis using various machine learning models.

TABLE I. SUMMARY OF RELATED WORKS

Ref.	ML Tools	ML Models	Evaluation Metrics	Applications	Open Issues Discussed	Future Directions
[4]	√	√	√	×	×	√
[5]	√	√	√	×	×	√
[6]	√	√	√	×	×	√
[7]	×	√	×	√	×	√
[1]	√	√	×	√	√	√
[8]	√	√	√	√	×	√
[12]	√	√	√	√	√	√
[9]	√	√	×	√	×	√
[10]	√	√	√	×	√	√
[11]	×	√	√	×	×	×
Ours	√	√	√	√	√	√

The contribution of the paper is as follows:

- Comprehensive review of state-of-the-art ML models for SA.
- Discusses various tools and evaluation metrics used in sentiment analysis models.
- Identification of diverse applications of SA.
- Analysis of limitations and potential future research directions.

The organization of the remaining paper is as follows: Section II gives the background of ML and DL. Section III presents ML models used for SA. Section IV presents the applications of SA. Section V discusses the statistical analysis and evaluation measures. Moreover, Open Issues in SA using ML and future research directions are discussed in section VI. Finally, Section VII concludes the paper.

II. METHODS: STATE OF THE ART

Before exploring the intricacies of tools and techniques, it is essential to establish a foundational comprehension of both machine learning and deep learning. Machine learning is defined as "the empirical study of algorithms and computational models on computers, utilizing prior experiences to continuously enhance performance in a specific task or make accurate predictions" [13]. The term 'experience' encompasses diverse datasets embodying the complexities, patterns, and nuances inherent in the problem domain. The key principle is that these algorithms and models are not explicitly programmed but are designed to learn and adapt from historical information. This data-driven approach enables machines to discern underlying structures, relationships, and trends, allowing them to generalize patterns and make informed predictions when faced with new, unseen data [40]. The process involves training machine learning models on historical datasets, enabling them to identify and internalize patterns that can later be applied to novel instances. This iterative learning loop promotes continuous improvement, as models refine their predictions over time based on feedback and exposure to additional data.

Deep learning (DL) is a machine learning approach built upon the idea of algorithmic learning for data representation, employing artificial neural and belief networks constructed with multiple layers of modules. The input undergoes analysis and classification, with the output from one layer serving as the input for the next layer, creating a sequential flow of information. In a nutshell, backpropagation [14] is a technique that trains deep learning models by calculating how much each part of the system contributes to errors [15]. No matter what kind of data the model receives (sound, pictures, or text), it gets turned into numbers and grouped into categories. Each layer in the system builds on the one below it, finding more complex patterns from simpler ones. This layered structure is like a series of nested boxes, with each box representing a more abstract idea. Deep learning can also be compared to multi-step computer programs, where each layer acts like the computer's memory after it's done a specific set of tasks. The more layers a deep learning model has, the better it gets at picking out and remembering important information.

III. MACHINE LEARNING MODELS IN SENTIMENT ANALYSIS

There are several ML models that have been used for sentiment analysis. This section discusses different ML models used in SA.

- *Random Forest* – A machine learning algorithm known as a trademark of Leo Breiman and Adele Cutler is widely used. This algorithm, which combines the outputs of multiple decision trees to produce a single result, has gained popularity due to its ease of use and flexibility in handling both classification and regression problems.
- In the context of classifying tweets as positive, negative, or neutral, the SVM and Random Forest (RF) algorithms were employed. In a study comparing different classification algorithms [17], random forest (RF) excelled at identifying positive data points. However, for negative and neutral data, support vector machines (SVM) generally came out on top. This suggests that RF might be a better choice when you're especially interested in correctly classifying positive examples,

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while SVM could be more suitable for handling negative and neutral data accurately.

- *Logistic Regression* – The logistic regression (LR) is categorized as a linear classifier and bears some resemblance to polynomial and linear regression methods. It is known for its speed, simplicity, and ease of result interpretation.
  Despite being primarily designed for binary classification, LR can also be extended to handle multiclass problems. LR employs various optimization functions such as lbfgs, liblinear, and different parameters like "C," "penalty," and "dual." The optimal values obtained were "C" set at 1.0, "penalty" as l2, and "dual" as false. In the evaluation [17]. LR demonstrated favourable results and proved to be well-suited for use with cross-validation techniques [18].
- *Decision Tree* – The decision tree (DT) adopts a decision tree model, resembling a tree-like structure to make decisions. Implementing DT does not necessitate scaling the data, and it has the capability to implicitly conduct variable screening or feature selection. While decision trees (DT) are powerful tools, they can be prone to overfitting [14]. This means they might learn the training data too well and struggle with new, unseen data, leading to lower accuracy. When building a decision tree, two methods called the Gini index and information gain are used to decide how to split the data into different branches. These methods help the tree identify the most important factors for making accurate classifications.
- *Support Vector Machine* – The SVM has been extensively employed across various applications for solving classification problems [19]. It operates as a classifier by establishing a hyperplane to separate the data. In simpler terms, suppose you're trying to categorize things into two groups based on their features. Support vector machines (SVM) excel at finding a clear dividing line, like a straight line on a map, that best separates the two groups. This line is called a hyperplane in higher dimensions. To achieve this separation, SVM considers various factors like the penalty for misclassifications, a margin of error, and how similar data points should be considered. In this particular case, the researchers tested different settings for these factors (cost, epsilon, gamma) and found the best combination to be a cost of 4, a tiny margin of error (0.001), a setting called "scale" for another factor, and a basic straight-line separator (linear kernel function).

  In the realm of sentiment analysis, Support Vector Machines (SVM) emerge as a champion, particularly when it comes to deciphering the often-subtle nuances of negative and neutral data. This prowess is evident in its impressive accuracy of 67.61%, often surpassing other algorithms in these categories [14]. But SVM's strength extends beyond mere accuracy. Unlike some finicky methods that falter with large datasets, SVM maintains its peak performance regardless of data volume [15]. This scalability makes it a versatile tool for tackling diverse tasks [17]. Furthermore, SVM exhibits a remarkable adaptability to different languages, as showcased by its stellar accuracy of 91% in datasets using Arabic and Turkish [18]. Whether facing large volumes of data or navigating the complexities of foreign tongues, SVM consistently stands out as a reliable and accurate tool for sentiment analysis [20].
- Naïve Bayes – The Naive Bayes classifier operates as a straightforward generative model, learning the probability distribution to predict conditional probabilities. It utilizes Bayes' Equation, where $P(x1,\dots, xn|y) = \prod_{i=1}^{n} P(yi|x)$. In this equation, x represents the event for probability prediction, y is the new evidence related to x, $P(x|y)$ is the posterior probability, $P(y|x)$ is the likelihood, $P(x)$ is the prior probability, and $P(y)$ denotes the evidence. The Multinomial Naive Bayes (MNB) algorithm, a classic variant of Naive Bayes, handles data distribution multinomially and finds widespread use in text classification within Natural Language Processing (NLP).When it comes to speed, Naïve Bayes sprints ahead of the pack [2]. It boasts the fastest runtime among the algorithms considered, making it an attractive option for tasks where quick analysis is crucial. But speed isn't its only strong suit. Naïve Bayes also demonstrates impressive adaptability, holding its own in terms of accuracy regardless of the feature extraction technique used. Whether analyzing words based on Term Frequency-Inverse Document Frequency (TF-IDF) or Count Vector (CV), its accuracy score hovers consistently within the range of 0.66 to 0.73 [15]. This stability means you can rely on Naïve Bayes to deliver dependable results without getting bogged down in the specific details of your data preparation.
- *K-Nearest Neighbors* – The KNN is applicable to both classification and regression problems, relying on the notion that similar values are situated in proximity [16] [21]. It calculates similarities between input sample datasets. Finding the perfect number of neighbors is key for a K-nearest neighbors (KNN) classifier to make good predictions. The researchers tested different neighbor settings between 1 and 30 using a technique called the Elbow method. After some trial and error, they discovered that a KNN model with a specific number of neighbors (the optimal number isn't mentioned here but was likely identified through the Elbow method) achieved the best results. This winning model scored an impressive 0.73 on a performance metric called F1-score, which was validated using a reliable technique called 10-fold cross-validation [14].

IV. Applications Of Sentiment Analysis

Sentiment analysis has a wide range of applications in different fields. Some of the prominent applications of sentiment analysis are shown in Fig 2 and discussed below.

A. **Business Intelligence:** SA is a game-changer for businesses. It lets them sift through massive amounts of data quickly and cheaply. This gives them a clear picture of what their customers are really thinking, whether it's praise or complaints. This information is a goldmine for businesses of all kinds. They can use it to analyze their

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competition, understand market trends, and ultimately make smarter decisions that save time and money. It's a win-win for businesses and consumers alike [14].

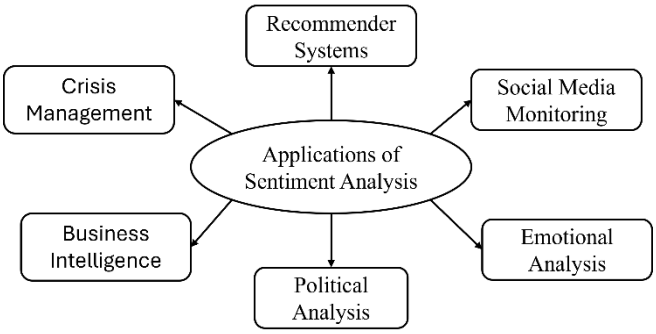


Fig 2. Applications of Sentiment Analysis.

B. **Political Analysis:** Sentiment analysis is a powerful tool used in political analysis to gauge what the public thinks about political candidates, parties, policies, and current events. It comes in handy for political campaigns as they can use this information to customize their messages and strategies to better connect with voters.

C. **Emotional Analysis:** Sentiment analysis goes beyond just identifying positive or negative feelings; it can also pinpoint specific emotions conveyed in written content. With the help of advanced natural language processing techniques, sentiment analysis models can categorize text into different emotional states like happiness, sadness, anger, fear, or surprise[22].

D. **Social Media Monitoring:** Businesses use sentiment analysis to keep an eye on social media platforms. They want to know what the public thinks about their brand, products, or services. It helps them understand how customers feel, spot new trends, and quickly address any concerns that come up [23].

E. **Recommender Systems:** Sentiment Aware Recommender Systems have been developed by researchers to incorporate users sentiment into recommendation process [24], [25].

F. **Crisis Management:** When things go haywire or emergencies strike, sentiment analysis can be a total game-changer for government agencies, emergency services, and non-profit organizations. By keeping a close eye on social media in real-time, they can quickly spot updates, figure out which areas are in distress or need help, and coordinate their response efforts more efficiently [27].

V. Statistical Analysis and Evaluation

This section discusses different tools and evaluation metrics used to measure the performance of sentiment analysis models. This will help the naïve researchers who want to explore sentiment analysis to find the suitable tool to work and evaluate metric for measuring the effectiveness of their models. Table 2 provides a summary of sentiment analysis tools. NLTK is a Python platform for language data manipulation [17], the NRC Lexicon offers a comprehensive emotional association dataset [20], VADER utilizes a rule-based model for sentiment analysis [2][26][25], and WST is a specialized tool for automated web data extraction [27].

TABLE II. Different tools used for SA using ML

Tool name	Open access	Description	Reference
Natural Language Tool Kit (NLTK)	Yes	a prominent platform for writing Python programs tailored for human language data manipulation	[17]
NRC Word-Emotion Association Lexicon	Yes	This lexicon is highly renowned and extensively utilized for its proficiency in sentiment analysis. Comprising 27,000 words, each word is linked to 10 values, encompassing positivity, negativity, and eight additional emotions. These additional emotions consist of fear, anger, anticipation, trust, surprise, sadness, disgust, and joy	[2]
Valence Aware Dictionary for Sentiment Reasoning (VADER)	Yes	VADER operates on a rule-based model utilizing a valence-based lexicon, which not only determines the polarity of sentiment but also expresses its intensity. It's important to note that the VADER lexicon is based on the general population's use of the English language	[28][29][27]
Web Scraping Tool (WST)	Yes	Specialized tool designed to automatically extract data from websites. It proves valuable for individuals' seeking information from databases in various ways	[30]

Evaluation Metrics – Evaluation and comparison with state of the art (SOTA) approach is an import aspect of any research [31]. SA approaches are evaluated using different evaluation measures. Table 3 shows different evaluation metrics used in reviewed papers. Some of the extensively used evaluation metrics are discussed below.

- *Accuracy:* The proportion of correctly predicted sentiments out of the total instances. The mathematical equation for computing accuracy is given as.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

- *Precision:* The proportion of true positive predictions out of all positive predictions. Mathematically, it can be written as.

$$Precision = \frac{TP}{TP+FP} \quad (2)$$

Recall: The proportion of true positive predictions out of all actual positive instances. It is particularly important in situations where missing positive instances (false negatives) are costly or have significant consequences. Mathematically, recall can be realized using the following equation.

$$Recall = \frac{TP}{TP+FN} \quad (3)$$

- *F1 Score:* The harmonic means of precision and recall, providing a balanced measure. The F1 score is especially

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useful in situations where there is an imbalance between the number of instances in different classes. Mathematical notation for F-1 score is given by following equation.

$$F-1\ score = \frac{2*Precision*Recall}{Precsion+Recall} \quad (4)$$

- Mean Squared Error (MSE): It is applicable in regression-based sentiment analysis tasks, MSE measures the average squared difference between predicted and actual sentiment scores. The mathematical equation for MSE is given as follows.

$$MSE = \sqrt[2]{\sum_{i=1}^{N}(x_i - \bar{x})^2} \quad (5)$$

TABLE III. EVALUATION METRICS

Ref	Accuracy	Precision	Recall	F_score	MSE
[13]	√	√	√	√	×
[17]	√	√	√	√	×
[22]	√	√	×	×	×
[23]	√	×	×	×	×
[25]	√	×	×	×	×
[26]	√	√	√	√	×
[30]	√	×	×	×	×
[32]	√	×	×	√	×
[33]	√	×	×	×	×
[34]	√	√	√	√	√
[35]	√	×	×	√	×
[36]	√	×	×	×	×
[37]	√	√	√	√	√
[38]	√	×	√	√	×
[39]	×	√	√	×	×

VI. OPEN ISSUES AND FUTURE DIRECTION

A. Some major open issues that has been identified in reviewed papers on SA are.

- Linguistic barrier while doing the SA, sentimental analysis also gives wrong classification if the reviews convey tones of sarcasm or irony. [17]
- At times, figures and graphs might contain data that is not available in the article's text, which will be ignored with the use of current models. [41]
- Noisy and inappropriate features in text are ignored. [38]
- Fake accounts can reduce the performance of sentiment classification, sentiment visualization showed a minor influence from fake accounts. [23]
- NLP challenges often cause hindrance in efficiency and accuracy of sentiment analysis. [24]
- Considering emoticons along with text can greatly influence sentiment analysis, this 'influence' is yet to be quantified. [42]
- Extracting meaningful information from short text messages is challenging. - Analyzing texts with mixed languages and short length is difficult. [3]
- Customization of dataset for multi-categorical classification is time-consuming. Performance of classifiers can vary with changing domain and dataset size. Building a well-defined large dataset is laborious and demanding in terms of time and effort. [39]

B. Future directions for these problems could be-

- Developing sentiment analysis models that are more vigorous to linguistic variations, dialects, and cultural nuances.
- Develop models that effectively extract information from textual as well as non-textual elements (e.g., graphs, figures) in a document.
- Exploring advanced filtering techniques that can automatically identify and exclude noisy or inappropriate features from the text.
- Develop algorithms to identify and then filter out content generated by fake accounts, such as using user-behaviour analysis and network analysis.
- Conducting research to address NLP issues such as word obscurity, context understanding, and morphological complexity, that can impact sentiment analysis.
- Operating studies to quantify the impact of emoticons on sentiment analysis and developing models that can effectively comprise of emoticon information into the analysis.
- Develop models especially designed for short text messages, utilizing techniques such as attention mechanisms and context-aware embeddings.
- Discovering methods to automate the customization of sentiment analysis models for multi-categorical classification to reduce the time-consumption.

VII. CONCLUSION

Sentiment analysis, powered by machine learning, has become a valuable tool for extracting insights from diverse data sources. In recent years, several studies have been published that uses various machine learning methods for sentiment analysis. However, it is difficult to find all these studies to understand the background, concepts, limitations and possible future directions. Therefore, this paper summarizes and highlights the potential of SA across various applications while acknowledging its limitations. By addressing open issues and exploring future research directions, SA can evolve into a more robust and versatile tool for understanding public opinion, analyzing trends, and making informed decisions in various fields.

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